Sagar Patel

Software Validation Engineer | Systems & AI Engineer

Riverside, CA, USA | sagar@myjobemails.com | Open to Physical AI, Forward Deployed, Embedded Software, and Gen AI roles

SUMMARY

At Microsoft, I validate firmware for AI platforms through software-driven verification, platform bring-up, BIOS validation, and production-readiness testing. Previously at Ciena, I engineered embedded Linux networking firmware and Jetson Orin inference pipelines; at Cisco, I built P4-programmable data-plane pipelines and AI traffic intelligence on BlueField DPUs; and at TCS, I developed bare-metal RTOS firmware for ARM telecom edge nodes. On the side, I build Watchpoint, StepAhead, FieldFix, SignalForge, HydraSwarm, and XG1.

CORE SKILLS

Platform Validation

Firmware Validation | Software-Driven Verification | Platform Bring-Up | BIOS Validation | Production Readiness Testing

Programming

C | C++ | Python | Bash | TypeScript

Embedded

Embedded Linux | RTOS | Yocto | Buildroot | Device Drivers | Secure Boot

Networking

P4 | DPDK | DMA Optimization | TCP/IP | BlueField DPU | Telemetry

Edge AI

CUDA | TensorRT | DeepStream | ONNX | Jetson Orin | Nsight Systems

Robotics

ROS2 | MuJoCo | Physical AI | Control Systems | Sensor Fusion

Web / Infra

Next.js | FastAPI | PostgreSQL | Docker | CI/CD | GitHub Actions

ATS KEYWORDS

Firmware Validation, Software Verification, Platform Bring-Up, BIOS Validation, Production Readiness, AI Platforms, Embedded Linux, Yocto, Buildroot, RTOS, Device Drivers, C, C++, Python, CUDA, TensorRT, DeepStream, Jetson Orin, BlueField DPU, P4, DOCA, ROS2, MuJoCo, FastAPI, Next.js, PostgreSQL, Graph/RAG systems, CI/CD, secure boot, MISRA-C, high-performance networking, edge AI, multi-agent AI

PROFESSIONAL EXPERIENCE

Software Validation Engineer - Microsoft

Sep 2026 - Present | Hillsboro, OR, USA

Firmware validation for Microsoft AI platforms across platform bring-up, BIOS validation, software-driven verification, and production readiness.

- Validate firmware for Microsoft AI platforms through software-driven verification, platform bring-up, BIOS validation, and production-readiness testing.

Senior Embedded Software Engineer - Ciena

May 2024 - Aug 2026 | Remote, USA

Embedded Linux networking firmware for logistics edge gateways and edge AI inference on NVIDIA Jetson Orin.

- Optimized packet processing, DMA transfer paths, and interrupt handling to improve sustained throughput by 31% during peak fulfillment operations.
- Led Yocto-based platform work including BSP upgrades, secure boot integration, CI validation, and release management, reducing bring-up and deployment timelines by 24%.
- Built CUDA and TensorRT inference pipelines on Jetson Orin to detect conveyor and sorter anomalies in real time, reducing unplanned downtime by 22%.

Embedded Software Engineer - Cisco Systems

Jan 2023 - Apr 2024 | Austin, TX, USA

Switching firmware with P4-programmable data-plane pipelines and AI-assisted traffic intelligence on NVIDIA BlueField DPUs.

- Integrated P4-programmable pipelines with switching ASIC SDKs, improving forwarding stability and reducing packet loss by 31% in large-scale validation.
- Implemented NVIDIA BlueField DPU workflows with DOCA and ONNX inference to improve encrypted anomaly classification accuracy by 24%.
- Optimized DPU offload scheduling and inference workloads, reducing host CPU utilization by 29% while maintaining throughput under peak traffic.

Embedded Software Engineer - Tata Consultancy Services

Jul 2020 - Aug 2021 | Remote, India

Bare-metal RTOS firmware and embedded Linux for ARM-based telecom edge nodes in carrier-grade networking environments.

- Developed bare-metal RTOS firmware for ARM-based telecom edge nodes, implementing multicast packet replication and QoS shaping to boost reliable delivery by 28%.
- Wrote custom kernel modules and device drivers for high-speed Ethernet interfaces, cutting packet jitter by 35% in distributed deployments.
- Integrated bootloader development with firmware authentication and MISRA-C compliant coding, streamlining secure deployment across platforms.

SELECTED PROJECTS

Offline OTA Update System - Embedded Linux (2026)

Signed bundles, atomic staging, health-check promotion, rollback, and audit history for offline edge Linux devices.

- Offline delivery over USB/local HTTP
- Automatic rollback on health-check failure
- Local audit history and dashboard

Embodipedia - AI / Multi-Agent Systems (2026)

Self-maintaining encyclopedia with typed claims, perspective routing, citations, and temporal views.

- Canonical/bull/bear routing
- Cited prose with unverified badges
- Live ingest and revision history

XG1 - Physical AI / Robotics (2026)

Humanoid robot learning pipeline for Unitree G1 combining teleop, DeepLake data, and GR00T fine-tuning.

- Won 2 hackathon tracks
- Meta Quest 3 + MuJoCo teleop
- DeepLake streaming for fast training

EDUCATION

- M.S. in Computer Science, Sofia University, Oct 2025 - Jun 2027
- M.S. in Embedded and Cyber Physical Systems, University of California, Irvine, Sep 2021 - Dec 2022
- B.Tech, Charotar University of Science and Technology, Jul 2017 - Jun 2021

CERTIFICATIONS

- NVIDIA-Certified Associate: Generative AI LLMs (NCA-GENL)

CONTACT

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